

23CSE472 – Artificial Intelligence and Robotics

Lab Worksheet – 3

Quaternions

1. Quaternion Representation

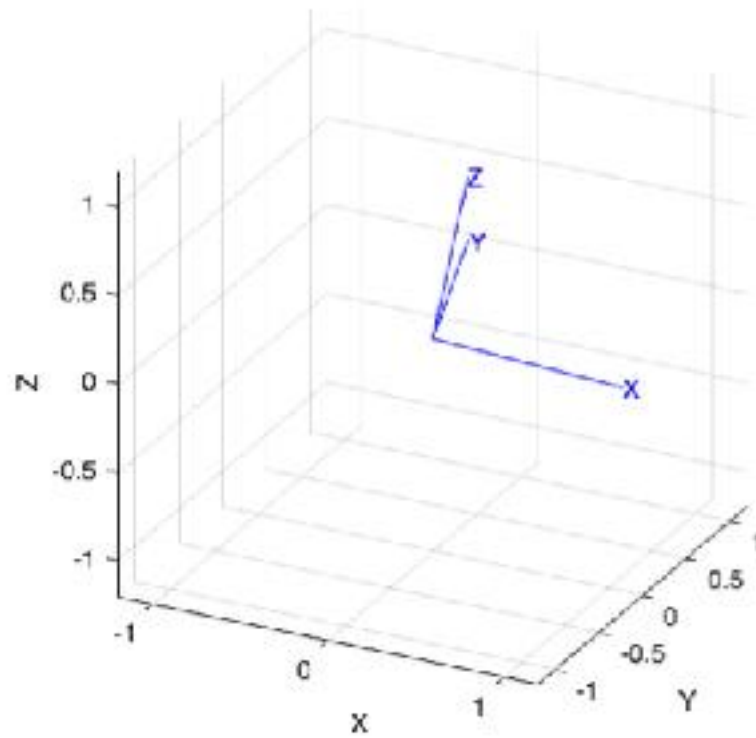
A quaternion is 4-element mathematical object comprising a scalar s , and a vector v

which can be considered as a pair (s,v) . In the Toolbox it is denoted by $q = s \ll vx, vy, vz \gg$.

A quaternion of unit length can be used to represent 3D orientation and is implemented by the subclass UnitQuaternion.

```
>> q = UnitQuaternion(rpy2tr(0.1,0.2,0.3))
q =
    0.98335 < 0.034271, 0.10602, 0.14357 >
>> q1 = inv(q)
q1 =
    0.98335 < -0.034271, -0.10602, -0.14357 >
>> q/q
ans =
    1 < 0, 0, 0 >
>> q*q1 ans =
    1 < 0, 0, 0 >
>> q.R()
ans =
    0.9363 -0.2751 0.2184
    0.2896 0.9564 -0.0370
   -0.1987 0.0978 0.9752
```

```
>>q.plot
```



```
>> Rx = rotx(90,'deg')
```

```
Rx =
```

```
1    0    0
0    0   -1
0    1    0
```

```
>> q=UnitQuaternion(Rx)
```

```
q =
```

```
0.70711 < 0.70711, 0, 0 >
```

```
>> q.plot('frame','A')
```

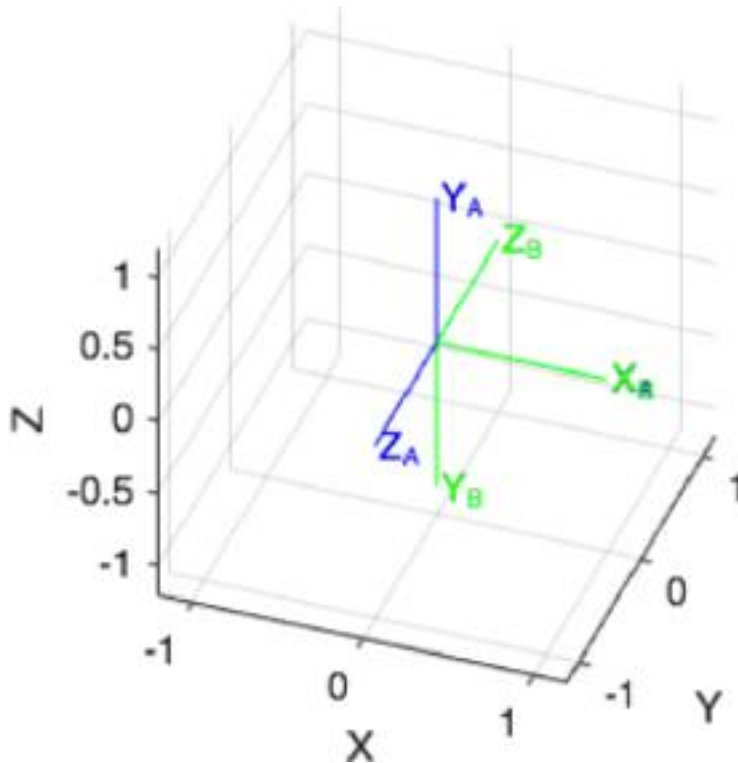
```
>> q1 = inv(q)
```

```
q1 =
```

```
0.70711 < -0.70711, 0, 0 >
```

```
>> hold on
```

```
>> q1.plot('frame','B','color','g')
```



```
>> q.angle(q1)

ans =

    1.5708
```

Try Out

1. If A is a rotation matrix obtained by rotating frame A by 45deg about X and 60deg about Y. if B is a rotation matrix obtained by rotating frame B by 60deg about Y and 45deg about X. prove $A \neq B$ by numerical and graphical representation. Hint: use “hold on” and “hold off” MATLAB function to plot two frames in a single plot.
2. Prove as in question 1 with:
 - i) A is a rotation matrix obtained by rotating frame A by 30deg about Z, then 80 deg about X and 126deg about Y. and
 - ii) B is a rotation matrix obtained by rotating frame B by 126deg about Y, then 80deg about X and 30deg about Y